

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

DTS 304 — Past Questions & Materials
by Renance

Course: DTS 304
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 300

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Paper 1 of 1 — DTS 304 Data Management I

Level 300 • First Semester • PastQuestion

Las FEDERAL UNIVERSITY OF PETROLEUM

= COLLEGE OF COMPUTING DEPARTMENT OF DATA SCIENCE

tp. its 1st Semester B.Sc. Examination 2025/2026

Course: DTS 304 Data Management I

Instruction: Attempt question

Time allowed: 2 hours, 30 m

1! and three other questions

notes

Question 1 (25 marks)

a. Define the following terms clearly with relevant examples:

i. Data (2 marks)

ii. Information (2 marks)

lii. Database (2 marks)

iv. Database Management System (DBMS) (2 marks)

b. Explain five advantages of using a Database Management System over traditional file-based systems. (5 marks)

c. Describe the following database models:

1. Hierarchical Model (1 marks)

1. Network Model (1 marks)

11. Relational Model (1 marks) a y.

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d. Compare the relational model with the hierarchical model, stating at least four ; differences. Why is the relational model the most widely used today? (4 marks)

e. Define the following keys with examples:

1. Primary Key (1 marks) ®

ii. Foreign Key (1 marks) 4

ii. Candidate Key (1 marks) .

iv. Composite Key (1 marks) ¥

f. Explain entity integrity and referential integrity with suitable examples. What problems E may arise if referential integrity is not enforced? (1 marks) - &

Question 2 (15 marks) be

Given the table:

SALES (SaleID, ProductID, SaleDate, Amount)

a. Write an SQL statement to create an index on ProductID.

b. Expl

(5 marks)

ain how indexing Improves query performance (7 marks)

¢. State one disadvantage of excessive indexing. (3 marks)

Question 3 (15 marks)

a. Explain the role of a Database Administrator (DBA). (4 marks)

b. Discuss four responsibilities of a DBA. (8 marks)

c. Define database performance tuning. (3 marks)

Question 4 (15 marks)

Given the table:

ACCOUNT (AccountNo, CustomerName, Balance)

a. Write SQL statements to transfer \$50,000 from Account A to Account B using é

Transaction control statements (BEGIN, COMMIT, ROLLBACK). (10 marks) a yh

b. Explain what should happen if the system crashes before COMMIT is executed. (5 t ne marks) ie Nv

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Question 5 (15 marks)

a Explain the meaning of query processing in a relational database system. (4 marks) «

b. Describe the three major stages of query processing. (6 marks)

c. What is query optimization? (2 marks) /

d. Discuss four techniques used to optimize SQL queries. (3 marks) ~

Question 6 (15 marks)

a. Define concurrency control in a database system. (3 marks)

b. Explain the following problems that arise in concurrent transactions:

i. Lost update (3 marks)

ii. Dirty read (3 marks)

iii. Unrepeatable read (3 marks)

c. Describe how locking mechanisms help solve these problems. (3 marks)